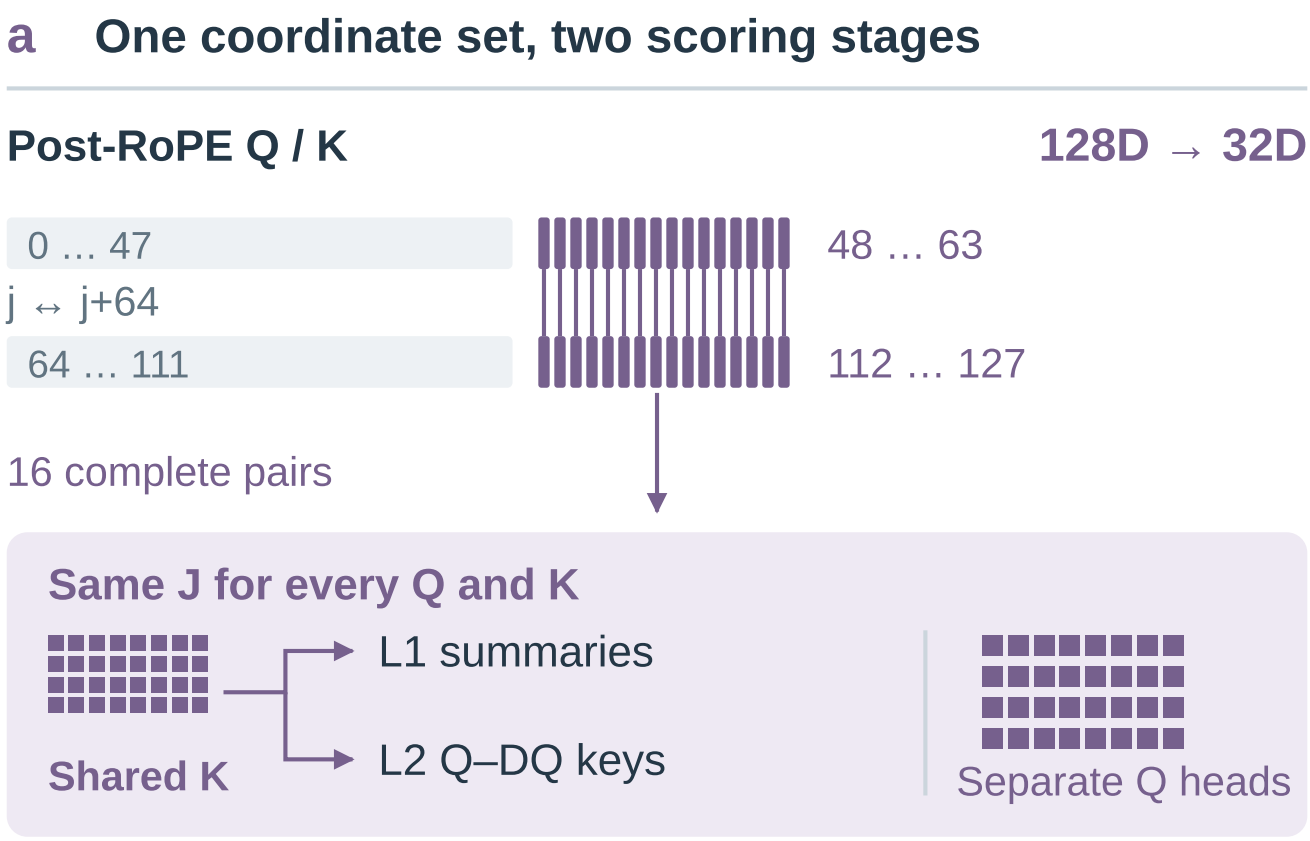
LoDA-SA: Supplementary mechanisms and historical evidence

Supplementary Figure S1: Shared scoring coordinates

Detailed views of subspace selection, regional allocation, and the gate lifecycle under the assumptions stated in the main text.



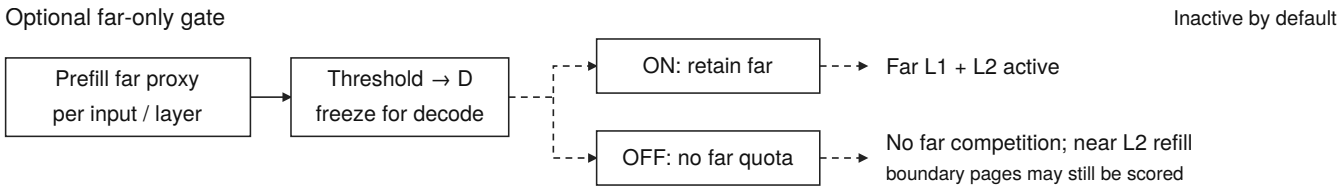
Subspace selection illustrated with the historical QDQ variant. The same 16 complete RoPE pairs feed L1 and L2 in both regions. This variant stores floating-point quantize–dequantize values, not packed INT4; QDQ is not required by the main-text bf16 configuration. Selected IDs define support over original 128-dimensional KV; physical access depends on the backend.

Supplementary Figure S2: Regional page allocation and single token pool



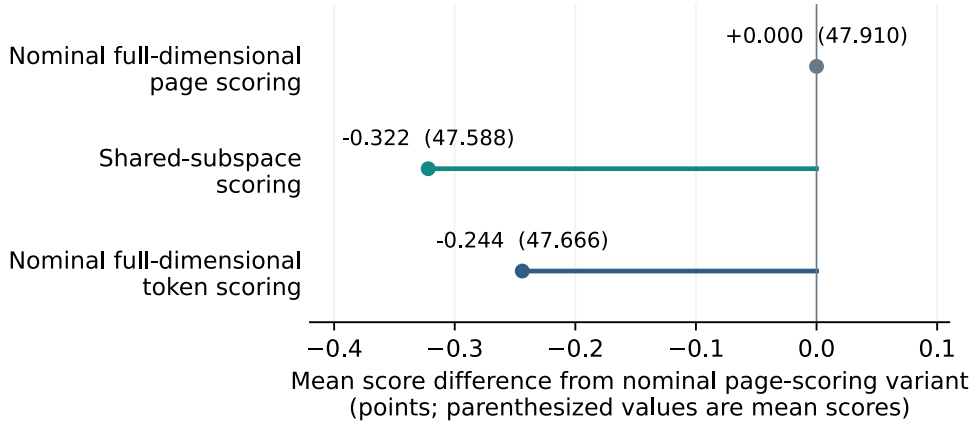
Regional selection and protected-token accounting. Under sufficient eligible capacity, $K_1 = 128$ gives 32 near and 96 far pages for the illustrative $(n_b, n_l) = (32, 4096)$ configuration with sink/SWA 4/128; the total 1024-token cap includes 132 protected tokens and one 892-slot free-competition pool. Up to four prefix positions can be reselected from their partial page, reducing unprotected coverage. The diagram's blocks are schematic. Per-query-head normalization precedes KV-group averaging. The measured decode runs halve the L1 cap to $K_1 = 64$ (32 near, 32 far pages); the diagram retains the $K_1 = 128$ layout of the measured RULER quality arms.

Supplementary Figure S3: Historical numeric-quota gate



Historical numeric-token-quota gate state transitions and conditional execution; this figure is not the current unified L2 algorithm. Prefill stores a decision; the first decode-path call freezes it. In the fixed numeric-quota path with valid length and nonempty far history, OFF disables far competition and final unprotected far selection; boundary bypass may still admit the sink-edge page to scoring. Only unused L2 quota refills eligible near tokens; the near L1 cap stays 16 pages in this alternate example. Global preparation and S-wide buffers/softmax remain. The control routes are schematic, not measured layer outcomes. The gate is disabled by default and has no matched quality–cost validation for the shared-subspace configuration.

Supplementary Figure S4: Descriptive configuration comparison



The five-task means are 47.910 (nominal full-dimensional page-scoring variant), 47.588 (shared-subspace configuration) and 47.666 (nominal full-dimensional token-scoring variant). The differences of 0.322 and 0.078 points from the shared-subspace configuration are descriptive. The nominal full-dimensional page-scoring variant lacks experimental metadata; the nominal full-dimensional token-scoring variant's L2-mode flag is not bound to an implementation revision that implements that mode. These labels therefore do not establish controlled stage interventions or a completed three-configuration ablation. Supplementary Note S7 gives the configuration evidence. None is dense FullKV, and no document-paired non-inferiority or matched quality–cost conclusion follows.

Supplementary Note S5: Gate observation and execution scope

The current n_l/n_b design disables gating. The following describes the historical numeric-token-quota extension, not a validated gate for unified L2. In unified L2, sink-boundary candidates can still yield far winners after ordinary far-page competition is removed. When the optional gate feature is disabled, ordinary far selection remains active; OFF denotes a calibrated per-layer skip decision only when the feature is enabled. The optional far-region gate observes a layer's prefill queries, stores a pending far-skip decision, freezes it at the first decode-path call, and reuses it until a new input clears the state (Supplementary Fig. S3). That call can still consume question tokens, so it need not see the complete question. Calibration uses at most the final 256 available query rows, full-dimensional scores and per-query-head softmax; its averaged far-attention weight sum is compared strictly with the threshold. The threshold controls a prefill observation; it does not bound far-attention weight sum or task error during later decoding.

Calibration geometry. For sufficient history, let S_c be the calibration length and $m_c = S_c - s - w$. The statistic sums attention over $[s, s + \lfloor(1-\alpha)m_c\rfloor)$ before averaging query rows and heads. The supported decode skip instead uses the historical page-aligned boundary Bb_n, where $L = \max(B, \lfloor\alpha(S - s - w)\rfloor)$ and $b_n = \max(\lfloor s/B\rfloor, \lfloor(S - w - L)/B\rfloor)$; its unprotected skipped interval begins at s. Even at equal lengths, these intervals need not coincide. With $B = 64$, $s = 4$, $w = 128$ and $\alpha = .25$, their right endpoints are 6049 and 6016 at $S_c = S = 8192$, but 703 and 704 at 1065. Thus the statistic is a proxy for far-region contribution, neither the exact mass of removed support nor a conservative bound on it. Protected prefix positions remain protected.

In the supported fixed-quota implementation (batch one, valid length and nonempty far history), OFF disables far competition and final unprotected far selection; it does not guarantee elimination of all far dot products, because a sink-edge page may remain eligible through boundary bypass. Unused L2 quota refills eligible near candidates without enlarging near's L1 page budget. Far-on, off-refill and off-no-refill are distinct interventions: removing far can change per-query softmax denominators and the aggregated near ranking. Consequently refill is not an isolated saving with an invariant near set. Preparation, quantization, repeat-to-float and S-wide buffers/softmax remain in the cost. The implementation does not support combining dynamic and static layer skipping. Batched state isolation, runtime state transitions and distribution-shift-triggered fallback have not been validated. Runtime adaptation of α, β, γ is outside the evaluated configuration.

This historical gate remains a separate candidate extension; its near-token refill rule does not apply to current unified L2. The archived gate task-level summaries do not identify generation-code revision, complete subspace flags or paired predictions, so they cannot validate this configuration. These implementation details qualify the mechanism rather than provide measured speed or quality gains.

Supplementary Note S6: Scope of the archived diagnostics

The following records retain the numerical detail behind the main-text limitations. They are author-archived diagnostics, not new experiments or matched validation of the shared-subspace method. Archived records carry ratio identities; these map to absolute quotas via $n_b = \max(1, \text{round}(\beta K_1))$ and $n_k = \min(\lfloor \gamma n_b B \rfloor, K_{\text{mid}})$, so $\beta = .0625$ at $K_1 = 128$ equals near quotas (8,320) and $\beta = .125$ equals (16,640).

Motivating diagnostics. Figure 1 uses archived Qwen3-8B observations with the following scoring and selection rules. Panel (a) retains the fixed HotpotQA0/L24/q8672 page arrays and saved selections from E140b (2026-10-23 figure revision; the earlier fixed MuSiQue1/L24/q14212 arrays and the retired top attention-mass strip remain in repository history): mass ranking and mean scoring each use 2017 tokens, and extrema scoring uses 2048. The fixed example is one of 3/128 observations that are extreme on both the page-count and missed-mass complementarity criteria (far share at the 87th percentile), not a median sample. Avg/minmax share 32 low-frequency coordinates and 32-head averaging; the reference uses full-dimensional probabilities. Regional allocations and realized token counts differ, precluding regional-specialization or matched-budget optimality claims. Among 132 mid pages, all sets contain 32 pages. Relative to mass-rank, avg misses 8 and minmax misses 9, including two missed by both; six are missed only by avg and seven only by minmax. Each scorer selects 8 or 9 pages outside the reference, respectively. Gray means neither selected nor a missed reference page. These differences are not errors or task-quality losses, and the whole-page reference is not a proven optimal oracle. The spike at position 832 has one token carrying 99.85% of the page's stored mass; it illustrates this query rather than a population-wide mechanism. Panel (a)'s regional statistics (2026-10-23 figure revision: the stacked-column inset was withdrawn with the panel-count reduction; the numbers remain text-cited here) recompute regional membership from absolute intervals $[q - d_{hi}, q - d_{lo} + 1)$ and exact region bounds, then divides each query's regional missed reference mass by that query's total reference-set mass and averages the resulting shares: block-avg misses .0843 near and .1699 far, minmax misses .0787 near and .0420 far (means over 128 correlated observations). Missed page counts average 3.98/3.89 (near/far) for avg and 5.27/3.77 for minmax, so the page-count and mass criteria rank the scorers differently. A separate near-budget scan (near pages by avg, far pages by minmax, total cap 2048, near budget 128–1536 in steps of 64) never beats global minmax selection: the best split misses .1345 at near budget 1408 versus .1208 for global minmax, so budget division is not decided on this plane. The earlier quota-arm panel (global cap 2048 versus far/near 1024/1024) is retired; its per-query records remain in the repository history. These are correlated query/layer observations from four source sequences, not independent documents. Panel (b) retains the eight-cell coordinate ratios and unpaired random choice; it neither isolates frequency nor evaluates production scoring. Panel (c) uses only the 16-by-36 matrix in the historical far-profile record: 24 layer medians below .01, and 12–32 low-proxy layers per sequence. Its producer uses the final sampled query, sums logits within a GQA group before softmax, and defines far as $[64, t - 2048)$. This historical proxy is not the historical optional gate statistic in S5; low values are not a count of safely skippable layers. The separate 10-profile skip summary is not used. The accompanying numerical inputs include per-query selections and source hashes.

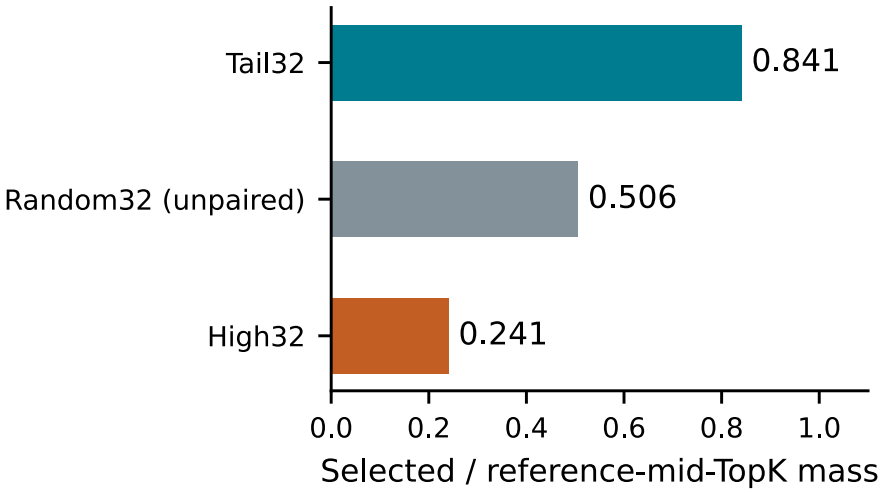


Figure 1: Secondary archived coordinate diagnostic: selected mass divided by full-dimensional reference mid-TopK mass, with middle cap 768 and separate 128/128 protection. The diagnostic page-average selector differs from production near-average/far-minmax scoring. Random32 is unpaired. No paired-random quality or output-error test is claimed.

Timing provenance. The main-text decode-TPOT figure uses the E152v4 formal record's clean arm (host pc40): Qwen3-8B bf16 on an NVIDIA H20-3e, Torch 2.8.0+cu128, greedy generation with fixed 128 output tokens, ABBA-interleaved arms with two processes per arm, and per-step device-synchronized timing that includes greedy argmax. The dense arm runs the native dense decode path on a dynamic cache without CUDA Graph (six measured repetitions per length); the sparse arm executes Ta-

ble I's geometry on a static cache with whole-step CUDA Graph replay (four repetitions), and a one-repetition eager reference without graph reports 99.025/99.893/134.063 ms. The graph arm's generated tokens match the eager reference bitwise at all three lengths. A co-tenant-polluted local replica (dense baseline inflated from 26 to 39 ms) passes the same bit-exact gate but is excluded from the reported means; warmup runs are excluded. An earlier E152v2 record reported medians 87.9/97.6/119.9 ms for an archived $(.125, .375, .625)$ dynamic-cache configuration without graph replay; E152v4 supersedes it as the figure source, and both records agree that no-graph sparse decoding is slower than dense. Backend routes are author-reported, not independently verified. The E127 5.47-fold figure compares old and optimized selector stage-time summaries, not dense FullKV. Neither it nor the E152 aggregates establishes a launch-bound cause. Source-data, collector and renderer hashes are supplied in `sources/evidence_bindings.json`.

Local selector timing and aggregation. The archived performance record reports two GPU-crossed rounds for a later implementation. Its associated source has a compact shared-subspace L2 path, distinct from the earlier LongBench implementation in Note S7; source inspection alone does not certify the timed execution. The per-round values are 1.3635/1.337 ms (nominal full-dimensional page-scoring variant), 1.3872/1.3324 ms (shared-subspace configuration), and 1.4794/1.4473 ms (nominal full-dimensional token-scoring variant). Its reported two-round medians are 1.3502, 1.3598 and 1.4634 ms, respectively. Midpoint medians computed from each pair are 1.35025, 1.35980 and 1.46335 ms; these are arithmetic recomputations, not additional measurements. The record does not supply per-stage medians or raw sequences and therefore does not establish the earlier description as a sum of five stage medians. The synthetic single-layer, batch-one, single-query 64K scope is distinct from complete layer or request execution and is not paired with the quality runs. The record describes 60 steps, 40 warmups and 20 steady samples, whereas the previously inspected harness implies 100 total and 60 steady samples. This discrepancy remains unresolved. The reported shared-subspace timing exceeds that of the nominal page-scoring variant; neither statistical significance nor complete-system speedup is established.

Regional and gate diagnostics. Protection-only scores 26.09 versus 47.91 with middle-history retrieval, but uses 256 rather than 1024 tokens, confounding allocation and budget. Gate calibration has 16 inputs with 36 layers each. Recorded far-attention proxies < 0.05 select 30 prefill and 22 decode records, with 13 shared. Across the prefill-selected records, decode proxies average 0.0582 and peak at 0.1678; each proxy is reported as the last-five-query mean, not a token-level maximum. These cross-phase changes establish neither quality preservation nor request savings. A separate author report gives five-task means 47.876 ($\epsilon = .05$) and 47.692 (.10), against reused 47.910: changes $-0.034/-0.218$. Each arm declines on three tasks; HotpotQA changes are $-1.05/-1.26$. Five-task bootstrap intervals $[-.584, .452]/[-.806, .400]$ are descriptive, not equivalence evidence. Execution records do not establish a matched shared-subspace comparison (S5). Matched three-policy quality and complete cost remain unverified.

Token-quota diagnostics. Archived E130/E98 summaries report retained full-dimensional causal attention mass, including protected tokens, rather than task accuracy. They average within each trace and then equally across 16 traces (14 LongBench and two constructed inputs). At $(\alpha, \beta) = (.25, .125)$, free token competition after L1 page selection yields 0.852, versus 0.8519 for the best tested γ (.875) and 0.827 for $\gamma = 0$, which assigns no near-token quota. Other grid cells have different valid-sample support, so their aggregate means require sample alignment before comparison. The inspected analyzer uses near-mean/far-extrema page scoring and 2048 middle-token slots plus 128/1024 sink/recent protection; this differs from the current compact protocol, and the historical execution has not been independently replayed. A separate E109 quota calculation treats zero far quota as insufficient to establish collapse. These diagnostics neither isolate a protective effect of regional token quotas nor rule out tuning gains; current-configuration free-competition task quality and complete-request cost remain unverified.

Archived absolute-quota grid (E155d). The committed summary reports 86 configurations plus one no-partition control on five Qwen3-8B LongBench tasks, one run per configuration. The top numeric-quota configuration $(n_b, n_k, n_l) = (32, 448, 4096)$ scores 48.668, quota-free $(32, 4096)$ scores 48.264, the best quota-free $(64, 8192)$ scores 48.300, and the control scores 47.462. Near-page marginal means are 46.371, 47.344, 47.965, 48.212 and 48.106 for $n_b = 4, 8, 16, 32, 64$, over unequal counts 12, 20, 24, 18 and 12; these pooled comparisons do not isolate a causal axis. The 69 numeric-quota-minus-quota-free comparisons span 17 geometries, average $+0.040$ points and are positive in 42 cases; at $(32, 4096)$, quota448 is $+0.404$ above quota-free. The three intervals in this JSON compare the top three numeric-quota arms, not quota-free selection. A later task record separately reports quota448-minus-off $+0.405$ with interval $[+.099, +.786]$, off-minus-control $+.803$ with interval $[-.194, +1.839]$, and quota448-minus-control $+1.209$ with interval $[+.177, +2.265]$; these later intervals lack independently verified raw replay and a confirmed document-cluster resampling protocol here, and do not establish current-method validation. The separately reported .933 top-half-page mass median was computed from all stored middle pages, including far pages, rather than a verified near-page subset; it cannot explain a near-region half-coverage rule. Current pooled L2 remains a design choice requiring matched input/configuration identities, quality and complete-request cost. The grid summary and later interval reports are distinct evidence sources.

Available evidence and missing controls. Matched dense FullKV quality, a full-dimensional sparse control, and request comparisons that include preparation remain missing; historical full-L1/tail-32 runs cannot fill them. At nominal 128K, the author-reported E146 RULER summary (cited in the main text) identifies Qwen3-8B with YaRN factor 4 and a sparse configuration with $\alpha = .125$, $\beta = .375$, $\gamma = .625$. Recomputing unweighted means from its 11 task scores gives 46.43 for FullKV and 23.98 for the sparse configuration, a difference of -22.45 points and a sparse/FullKV score ratio of 51.6%. This is a negative aggregate result,

not verified matched validation of the current n_l/n_b configuration with unified L2: raw predictions are absent from the inspected repository snapshot; paired input identities, scorer execution and the current compact execution identity were not independently verified. For the ordinary fixed-quota path without a quota override, the inspected formula gives $K_{2,\text{mid}} = 1024-128-128 = 768$ and $\text{round}(128 \cdot .375) \cdot 64 \cdot .625 = 1920$, hence a nominal far-token quota $\max(0, 768 - \min(1920, 768)) = 0$. This inference combines inspected code with reported configuration; the actual execution path has not been independently verified. It does not establish the sole cause of the score gap. The subsequently available 64K files contain aggregate scores and a manifest, not raw predictions; their receipt summaries omit fields including α, β and sink protection, so they do not close full run-configuration identity. No 64K scores are added to this comparison. These summaries do not isolate a causal effect of YaRN or far-region candidate starvation. The historical static-layer policy did not meet its improvement threshold. That negative result and the mixed-policy gate diagnostics constrain interpretation but do not directly test this candidate. Their scores and provenance remain in the evidence register. These Qwen3 diagnostics establish no cross-family or production benefit.

Supplementary Note S7: Budgets and configuration identity

Current configuration and historical identities. LoDA-SA uses explicit n_l/n_b, unified L2, 4/128 protection, dense prefill and sparse decode, with gating disabled; the illustrative, unvalidated compact point uses 32 near pages, a 4096-token near extent and a free-competition middle pool, with implementation defaults still 128/128. The E149 summary reports 4/128 protection but contains inconsistent page/token-quota labels; complete predictions and a reconciled execution identity are unavailable. At $K = 1024$, disjoint protection leaves 892 nominal pooled middle slots, not a 320/572 or 640/252 regional split; those splits belong to historical numeric-token-quota configurations. The partial 64-token prefix page can reselect four protected positions, so distinct unprotected tokens are not guaranteed. The source interface currently requires `tli_near_tokens=off` alongside explicit near length and pages; zero or omission does not select this branch, and this compatibility sentinel is not a tunable token budget. StreamingLLM motivates small sinks but also reports benefits from longer semantic prefixes on LongBench; it establishes neither four-token optimality for Qwen3 nor a 128-token window. Changing 128/128 to 4/128 releases 124 retrieval slots at fixed K, confounding sink-length and retrieval-capacity effects. Historical representation scores, timing and protection diagnostics retain their own settings, not 4/128 labels.

Integer boundaries and summary support. In the current explicit-n_l/n_b partitioned path, with $S = t + 1$ valid cached positions including the current token and excluding padding, let $M = \max(0, S - s - w)$, $L = \max(B, n_l)$ and

$$b_n = \max(\lfloor s/B \rfloor, \lfloor (S - w - L)/B \rfloor),$$
$$b_w = \max(b_n, \lceil S/B \rceil - \max(1, \lfloor w/B \rfloor)).$$

For sufficient history and disjoint protection, far and near L1 intervals before removing bypass pages are $[\lfloor s/B \rfloor, b_n)$ and $[b_n, b_w)$, respectively. The unified L2 interval is $[B\lfloor s/B \rfloor, S - w)$, with the combined page mask. Rounding and alignment can change near length from n_l. Summaries use full physical pages, not unprotected intersections: the partial prefix page includes protected keys, and the last near page can include recent-window keys at nonaligned S. Pooled L2 still stops at $S - w$. These rules exclude clustering, quota overrides and short-history branches.

Scarce retained support. In current unified selection, L1 page masking precedes L2 softmax. Pooled TopK clips counts to interval width, not positive support. If the clipped count exceeds that support, zero ties can select cached tokens from unselected pages; final selection does not reapply L1 eligibility. Fig. S2 assumes sufficient support and disjoint protection; partial-prefix reselection is separate. Occurrence and quality effects in the reported runs remain unverified.

Fair comparison. Comparisons match model, input IDs, generation limits and task scorer; equal-K sparse comparisons count protection within the final unique cap. Retain and report each method's native phase, layer and protection policies; policy-matched adapters are separate controlled comparisons. Internal page caps may differ, but construction, scoring, ranking and traffic count toward complete-request cost. Report realized token counts and selection granularity, and match hardware and precision for timing.

Shared-subspace configuration LongBench settings. The five experimental metadata files specify page size 64, L1 cap 128 pages, final cap 1024 tokens, compression ratio 4, tail coordinates, $\alpha = .25$, $\beta = .125$, $\gamma = .625$, near-average/far-minmax page scoring and 4-bit quantize–dequantize values. They specify shared KV-head selection, dense prefill and disabled static layer skip. All five execution records identify producer `35b39bd8e43cf478bb4f0ab64a8457f08d87c864`. For these non-projection runs, that tree gathers 32 coordinates for L1 but scatters quantize–dequantize values into zero-filled width-128 L2 keys and scores them with full-width queries. Thus 32/32 describes active coordinates, not both physical contraction widths. The later non-projection shared-subspace configuration path gathers width-32 Q/K for L2; the archived quality score does not directly validate that implementation. These implementation paths do not establish runtime execution or finite-precision ranking equivalence. The recorded tree also gives prefix protection $2 \times 64 = 128$, recent protection 128, and sparse decode on all Qwen3 attention layers. These code-derived fields do not certify an unrecorded modified worktree or an execution trace. The normal long-history allocation is 16/112 near/far pages and 640/128 middle tokens, plus 256 protected tokens. The 8192 candidate slots do not imply 8192 final attention tokens, nor bound L2 scoring: ordinary L2 scores the full history before masking. Realized per-step counts were not recorded.

The configured input cap is 31500 for these runs. Task sample counts and maximum generated tokens are: Qasper 200/128; HotpotQA 200/32; MuSiQue 200/32; GovReport 200/512; RepoBench-P 500/64. The cap is not an observed input length. The complete model/tokenizer revision, runtime precision, hardware identity and scorer revision are not established by these experimental metadata; they must be bound for a matched comparison. Document-cluster identity, a matched dense run and a predeclared task-quality tolerance are still required for non-inferiority testing.

Nominal full-dimensional page-scoring variant. Five legacy prediction files have no manifest or execution record. Filename/verdict records identify page 64, L1 cap 128, final cap 1024, compression ratio 4 and the same nominal α, β, γ. Effective input cap, full implementation revision, protection, phase and complete scorer settings are not verified. Current defaults cannot fill these gaps.

Nominal full-dimensional token-scoring variant. Five manifests contain the same listed budget settings and an explicit full L2 flag. Four receipts omit the implementation commit. The MuSiQue receipt identifies `60027fba3672f33744917c4f59960d0f131f8ab0`, but that tree does not implement the recorded L2-mode field. The code actually executed is therefore not established by that commit. The nominal label and score are retained; full-dimensional L2 execution is not certified.

Quest comparator. The official evaluation uses dense multi-token prefill, sparse token-by-token processing, dense first two layers, and page selection per query head. Its 1024-token budget with chunk size 16 differs internally from 128 pages of size 64 in our selector. A controlled comparison aligns final unique K and phase/layer policy, while reporting the page quotas and head-sharing distinction separately. A paper's native configuration and a controlled adapter must not be pooled into one result. No matched Quest or FullKV quality–request-cost result is claimed here.

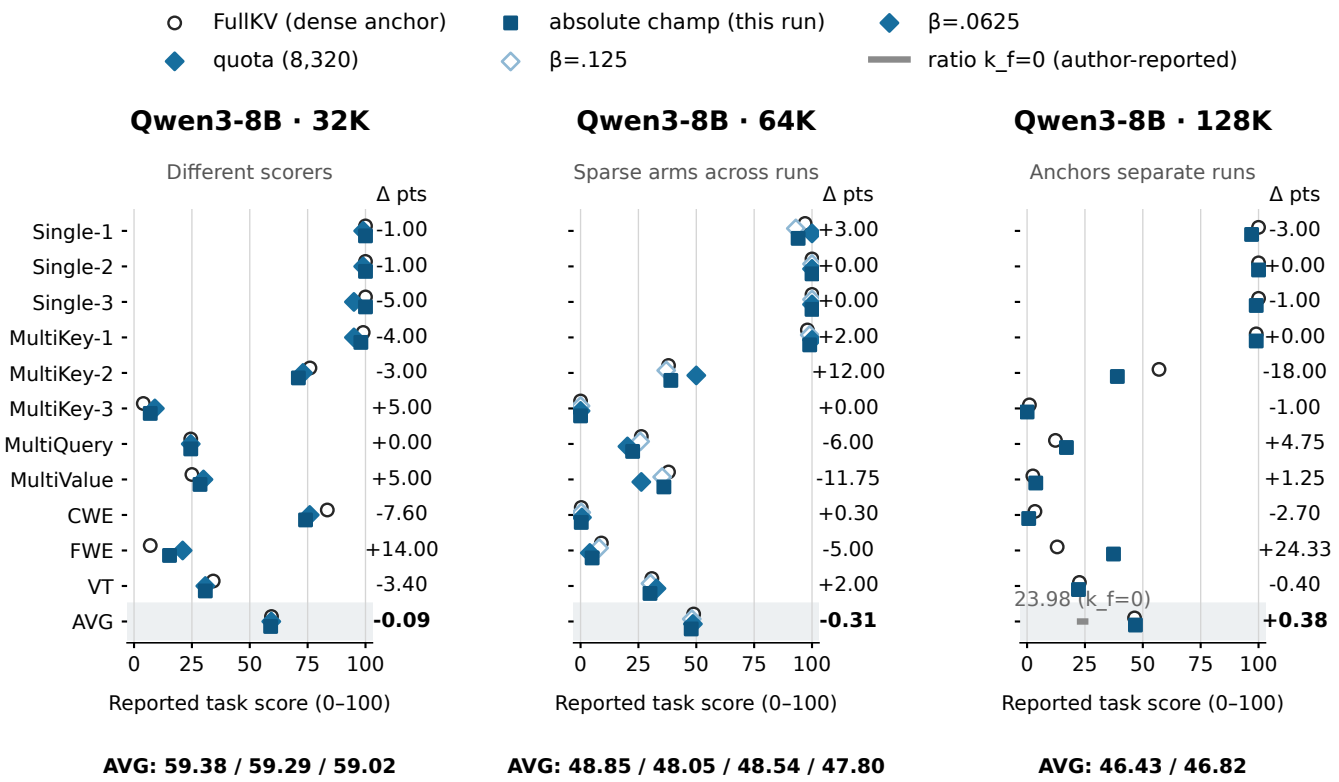
Other result families. The synthetic 64K selector timings are single-layer, batch-one, single-query summaries with an unresolved sampling protocol (S6); they are not bound to the LongBench requests. Gate summaries identify thresholds but not complete implementation and subspace settings. Protection-only uses 256 tokens versus its reference's 1024. Missing identity, unknown gate/refill settings, and unmeasured complete costs remain unknown rather than being inferred from current defaults.

Attribution controls (not completed here). Fixed-index replay would compare representation reuse and reconstruction on identical decode inputs and horizons, counting preparation and maintenance. Per step, match coordinate maps, summaries, quantization, selected token indices and final-attention computation between arms. Separately vary only regional/global L1 allocation, fixing page size, K_1, representation, protection, unified L2 and its total budget and K. Both arms use the same history-wide page scorer; report shortages and realized page/token counts. Competitors need not share K_1. Historical gate/refill comparisons are separate from the current unified-L2 design; a future gate study must first define and verify its intended support semantics.

The preceding designs are requirements for future matched tests, not experiments reported in the main text.

Representation maintenance and cost accounting. Appended keys, moving regional boundaries and coordinate-map changes require consistent derived views, logical token indices and region assignments. The shared-map identity does not prescribe an incremental maintenance algorithm. Quantize–dequantize keys use 16 floating-point levels, not packed INT4 arithmetic. Complete-request timing includes representation preparation, maintenance, calibration, selection, KV gathers, attention and synchronization, counting overlapping work once. TTFT ends at the first returned token; TPOT averages subsequent return intervals. These definitions are requirements for a matched evaluation, not a claim that complete-request costs were measured in the archived studies.

Supplementary Figure S8: RULER 32K–128K task scores



Historical numeric-token-quota scores plus one measured compact-design arm per length, for Qwen3-8B on 11 tasks, 100 examples per task and arm, at 32K, 64K and 128K. AVG is the primary statistic, an unweighted mean of all 11 task scores; task rows explain heterogeneity without imposing per-task zero loss. Differences are method minus dense FullKV in score points, without paired confidence intervals. The 32K archived method arm uses $\beta = .0625$. The 64K panel compares two sparse allocations, $\beta = .0625$ and $\beta = .125$, against a reused dense FullKV anchor. Reported sparse-arm labels share page size64, $K_1 = 128$, $K = 1024$, $\alpha = .25$, $\gamma = .625$, prefix4/recent128 and compression ratio4. Gate status, actual L1/L2 dimensions and realized selections are not independently established; these points must not be labeled validated shared-subspace-plus-gate results.

The 32K comparison spans machines. Task IDs, answer hashes, lengths and source-data hashes agree at the manifest level, but scorers differ and input identity is native versus legacy-partial. The -0.09-point AVG difference is an unmatched historical score difference. Neither panel establishes non-inferiority or a matched quality–cost benefit.

At 64K, the sparse arms, $\beta = .0625$ (E147h) and .125 (E146c), are reported on one machine; the E119 dense anchor is a separate run. Manifests align input IDs, answer hashes, lengths and source-data hashes across all three arms. Sparse scorer hashes match but producer-script and effective-configuration hashes differ; the dense anchor has legacy-partial input identity, a different scorer and only operator-declared YaRN settings. Their AVGs are 48.85, 48.05 and 48.54: sparse-arm difference $+0.80$, and $+0.31/-0.49$ relative to dense. All arms score zero on MultiKey-3 and below one on CWE. These descriptive aggregates have no reported paired intervals or replayed predictions; they do not isolate a change in β.

Sources: E147c sparse versus E116e historical FullKV (8B, 32K) and E147h/E146c sparse arms with the E119 FullKV anchor (64K) at revision `51b4679c52feebc5327775c660db41550b1c8782`. The editable figure directory contains the exact JSON/manifest snapshots, Git blob provenance, the score/difference CSV and reproduction script. These are aggregate recomputations plus the measured single runs below, not new paired experiments.

Measured compact-design runs (new single runs). One run per length executes the pre-halving Table I configuration string `tli_64_128_1024_c4_Aa0_b0_g1_sk4_sw128_np32_ntoff_nl4096`: page 64, K_1=128, K=1024, sink 4, window 128, near pages 32, free competition (`ntoff`), near extent 4096. The decode production runs of Q3 halve the L1 cap to K_1=64; the quality runs above predate that halving. Batch sizes are 8/2/2 with YaRN off/2.0/4.0 at 32K/64K/128K; receipts bind prediction SHA-256, producer script, seed 42 and generation identity per cell (11/11 tasks with receipts, no incomplete cells). AVGs are

59.02, 47.80 and 46.82 against separate-run dense anchors 59.38, 48.54 and 46.43 (gaps $-0.36/-0.74/+0.39$; 128K historical ratio arm with $k_f = 0$: 23.98). Per-task: NIAH-single-1/2/3 stay at 100/100/100, 94/100/100 and 97/100/99; MultiKey-3 is 7.0 at 32K and 0 at 64K/128K; CWE is 74.1/0.3/0.7 and MultiValue 28.5/36.0/3.75; FWE is 15.33/5.0/37.33. These are single runs without paired intervals; the dense anchors span generations, machines and scorers, so no non-inferiority follows. Result JSONs and receipts exist only in the locally untracked `exp/` tree: `t6_ruler_q8b_L32768/65536/131072_champ.json` plus per-length receipt JSONs; the E119 128K dense anchor is `e119_ruler128k_formal_fullkv.json`. The 23.98 ratio-interface aggregate has no per-task JSON and is drawn from the author-reported S6 record.

Ported-baseline comparison arms (new records, not shown in the figure). Ported 32K baselines: JSSA 58.7 (budget 1024, rank 32; basis calibration $8\times$ $\sim$32K is shorter than the vendor default 8×131072, disclosed), AsyncTLS 48.33 (nominal token budget 512; a semantic port of the paper protocol because official code is unavailable; the author describes this as a sorted-index rerun replacing 50.09; the sorting effect has not been independently verified), Quest 42.85 (native config) and FASA 18.64 (budget 1024, threshold 0.8, k16). Reported longer-context scores are 33.88 for AsyncTLS at 64K and 37.47 at 128K (both sorted-index reruns replacing the shuffled-index records), and 1.72/1.40 for FASA at 64K/128K. An eager-torch latency record of the same ports (single H20, synthetic needle prompts, 128 generated tokens) reports decode TPOT at 32/64/128K of 26.05/39.62/66.94 ms for dense, 33.30/39.68/61.69 for JSSA (undercutting dense at 128K), 75.64/133.00/265.91 for FASA, 105.29/123.42/160.25 for AsyncTLS and 177.04/317.38/621.37 for Quest. These eager implementations represent method semantics rather than official kernel costs and are paired with neither the task-quality runs nor the archived decode-TPOT figure. K2 budget rows on one machine: $K = 512$ scores 56.00 with CWE 32.5 (versus 75.9 at $K = 1024$, a 43.4-point drop) while two NIAH tasks each drop 1 point and FWE rises 1.33; $K = 384$ collapses to 15.00, without establishing a budget floor from these two tested points. The β curve's third point, $\beta = .03125$ at $K_1 = 128$ (far pages 124, far tokens 732), scores 58.07; $\beta = .0625$ lies between the original .125-step grid points, halving near pages (16→8) and raising the far token quota from 252 to 572 relative to the .125 arm. A same-machine 128K pair scores 45.73 ($\beta = .0625$) versus 46.02 ($\beta = .125$) with the 46.43 FullKV anchor from a separate run ($\beta = .0625$ ratio 98.49%). These scores are author-reported, not third-party replays, and lack paired intervals. The new AsyncTLS/FASA result JSONs, manifests and receipts exist only in the locally untracked `exp/` tree; the identifiers below do not establish repository artifact availability. Sources: `jssa_r32k_formal`, `e119_ruler32768_formal_asynctls_vendor_fix`, `e119_ruler32k_formal_quest_vendor`, `e119_ruler32768_formal_fasa_vendor_10182330`, `e119_ruler65536_formal_asynctls_vendor_64k`, `e119_ruler65536/131072_formal_fasa_vendor`, `e147j_k2512/k2384_r32k_formal`, `e147d_r32k_formal`, `e147h_r128k_formal`.

Historical cross-model records (outside the main-text scope). Author-reported Qwen3-32B RULER32K AVG is 50.64 sparse versus 52.22 dense (-1.58). Cross-machine Llama-3.1-8B arms report 83.88/86.19 at historical near page/token quotas (8,320)/(16,640). These descriptive records lack matched execution and paired uncertainty; they neither extend the current Qwen3-8B main-text scope nor validate its n_l/n_b unified-L2 configuration.